| Name: | | Centre/Index Number: | | Class: | |
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# DUNMAN HIGH SCHOOL
## Preliminary Examination
## Year 6

## H2 BIOLOGY — 9744/03

Paper 3 Long Structured and Free-response Questions

**21 September 2021**

**2 hours**

Additional Material: Answer Booklet

**READ THESE INSTRUCTIONS FIRST**

Write your centre number, index number, name and class at the top of this page.
Write in dark blue or black pen.
You may use an HB pencil for any diagrams or graphs.
Do not use staples, paper clips, glue or correction fluid.

**Section A**
Answer **all** questions in the spaces provided on the question paper.

**Section B**
Answer any **one** question in the spaces provided in the answer booklet.

The use of an approved scientific calculator is expected, where appropriate.
You may lose marks if you do not show your working or if you do not use appropriate units.

The number of marks is given in brackets [ ] at the end of each question or part question.

| For Examiner's Use | |
|---|---|
| **Section A** | |
| 1 | / 26 |
| 2 | / 14 |
| 3 | / 10 |
| **Section B** | |
| 4 / 5 | / 25 |
| Total | / 75 |

This document consists of **18** printed pages (including this cover page) and **0** blank page.

For Examiner's use

## Section A: Long Structured Questions (50 marks)

Answer **all** questions in this section.

### Question 1

Severe acute respiratory syndrome coronavirus 2 (SARS-CoV-2) causes the disease Covid-19.

The SARS-CoV-2 is a single-stranded RNA enveloped virus. Glycosylated S proteins covering the surface of SARS-CoV-2 bind to the host cell receptor ACE2, mediating viral cell entry. Fig 1.1 shows the structure of the SARS-CoV-2 virus.

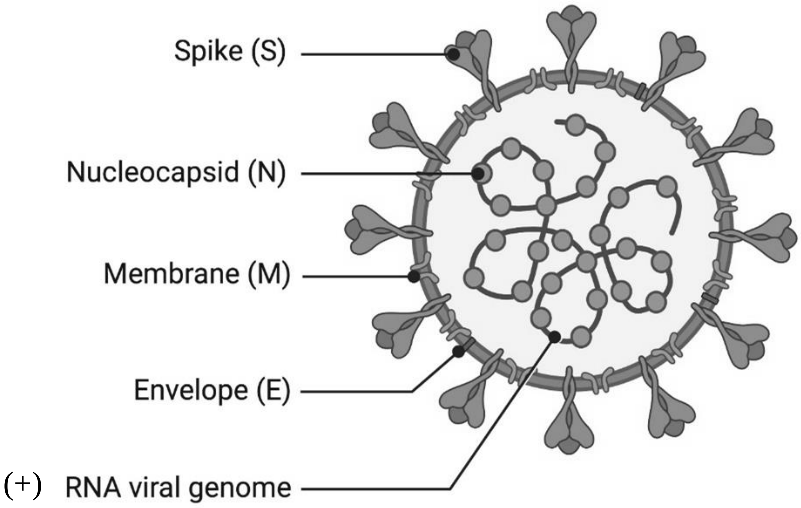


**Fig 1.1**

**(a)** With reference to Fig 1.1, compare the structure of the SARS-CoV-2 virus and the influenza virus.

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For Examiner's use

Fig 1.2 shows how the SARS-CoV-2 virus replicates.

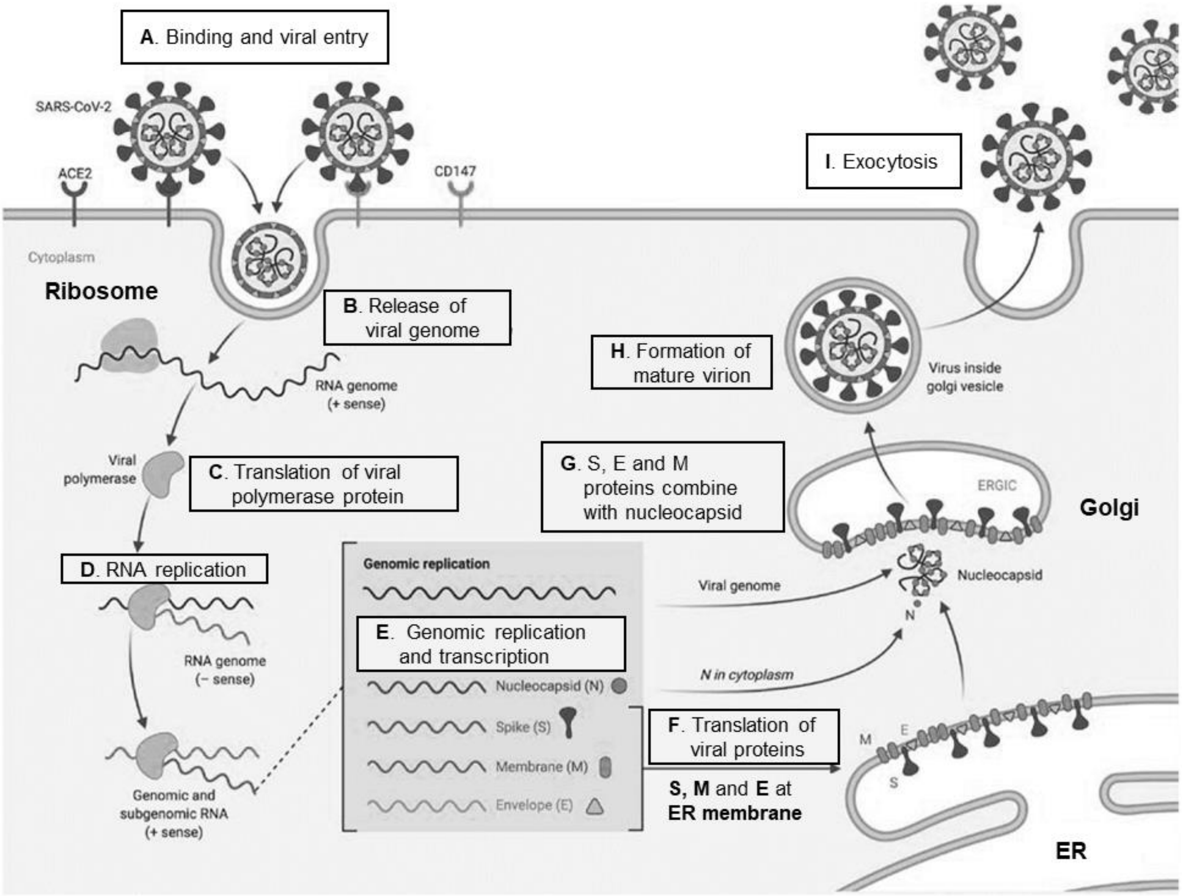


**Fig 1.2**

**(b)** With reference to Fig 1.2, explain why the SARS-CoV-2 virus is considered to be non-living.

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The SARS-CoV-2 genome is prone to mutation and has resulted in various variants. The B1617 variant of the SARS-CoV-2 virus has acquired mutations to the spike protein S1 unit that resulted in higher transmissibility rate. Fig 1.3 shows the mutations T478K and L452R on the S1 unit of the B1617 variant.

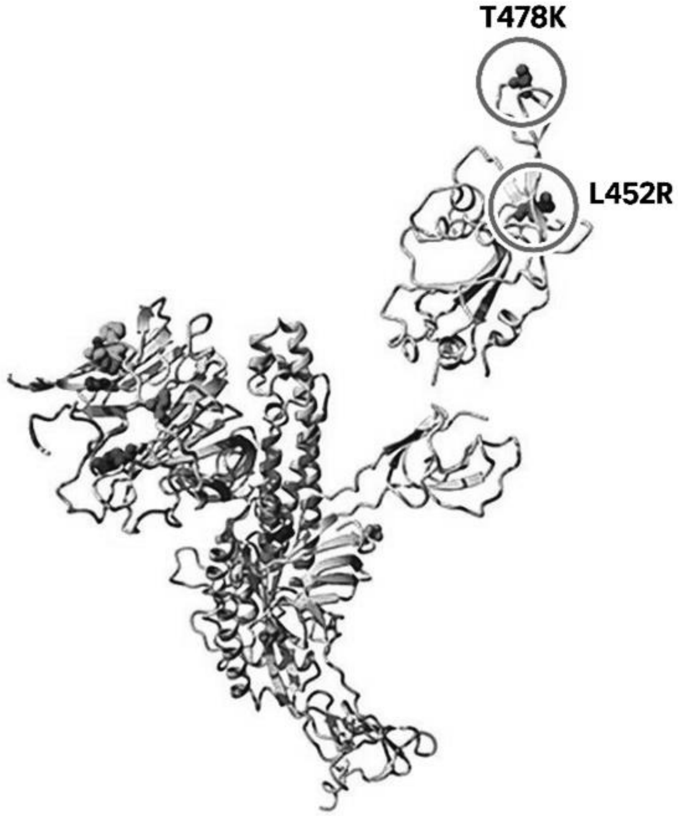


**Fig 1.3**

**(c)** Suggest why the SARS-CoV-2 genome is "*prone to mutation*".

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**(d)** Explain how mutations to the spike protein in the B1617 variant can result in "*higher transmissibility rate*" of the SARS-CoV-2 virus.

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For Examiner's use

For Covid-19, there has been a global initiative to share viral genome sequences with all scientists. Given a collection of sequences with sample dates, scientists are able to infer the evolutionary history of the samples in real-time as well as the history of transmissions. Fig 1.4 shows the evolutionary changes in the SARS-CoV-2 virus genomes between December 2019 to March 2020.

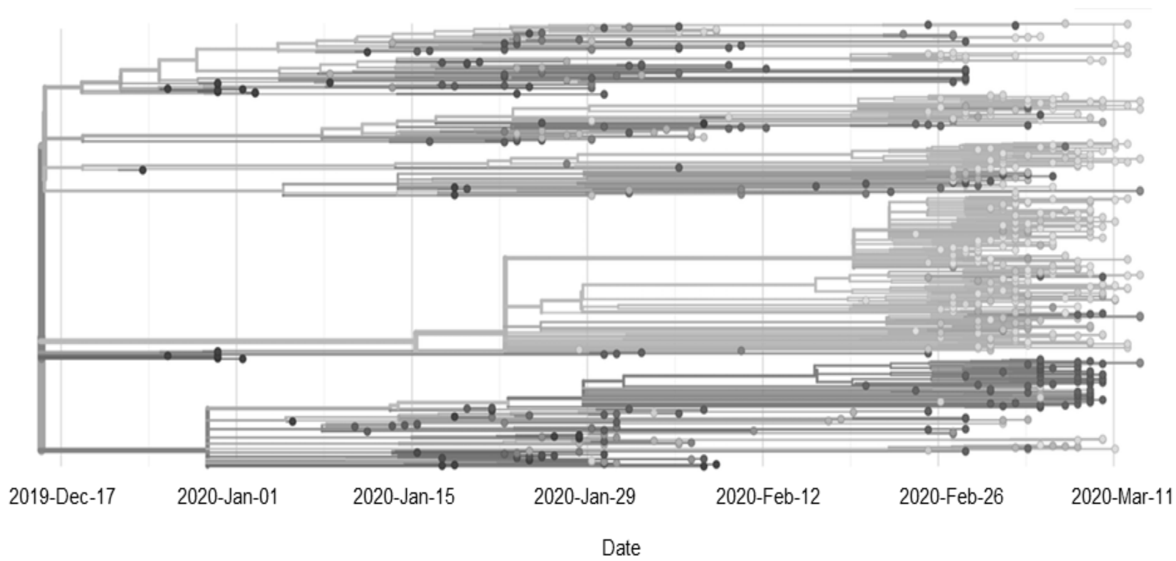


**Fig 1.4**

**(e)** Explain how having access to the SARS-CoV-2 genome sequences allows scientists to determine their evolutionary relationships.

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There are two main ways to test for infection with SARS-CoV-2. The first is a PCR test and the second is a antibody test using blood samples. Fig 1.5 shows a summary of how these 2 tests differ.

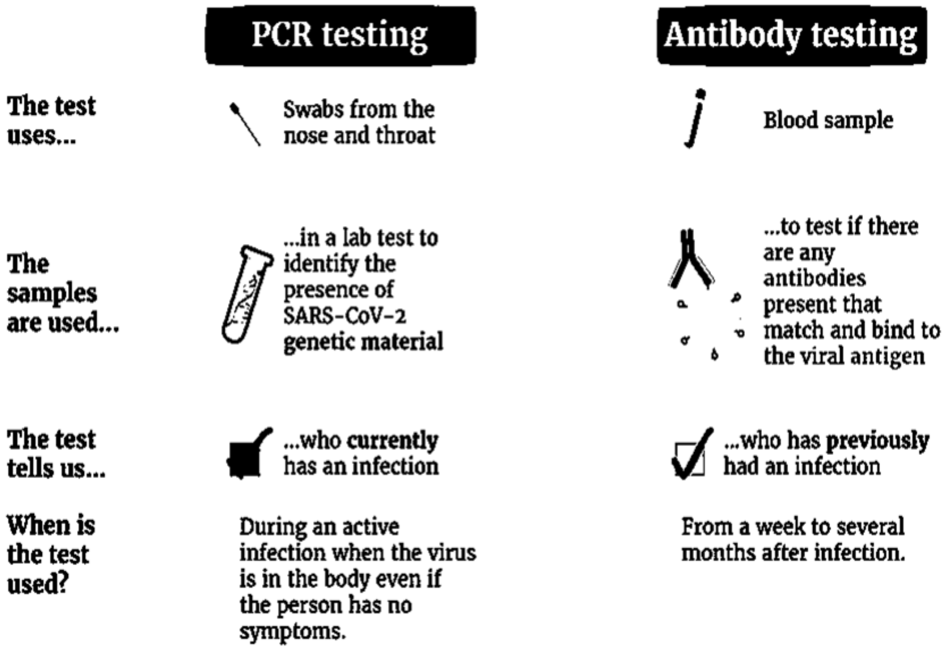


**Fig 1.5**

**(f)** In antibody testing, blood samples are tested for the presence of antibodies that are specific for the SARS-CoV-2 virus. However, this test cannot be used to determine if the individual is currently having a Covid-19 infection.

Explain why this is the case.

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For Examiner's use

To detect if someone is currently having a Covid-19 infection, a reverse transcription PCR test is carried out on nasal swab samples. The PCR test is very sensitive and accurate, and is able to detect as little as 1 virus particle from a swab sample.

**(g)** Explain how the PCR test is able to accurately detect the presence of the viral genome.

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Before a PCR test can be carried out, samples are first incubated with the enzyme reverse transcriptase as shown in Fig 1.6. After which, primers, enzymes and dNTPs are added into the samples and the samples are put through a series of temperature cycles to make more copies of the viral genome.

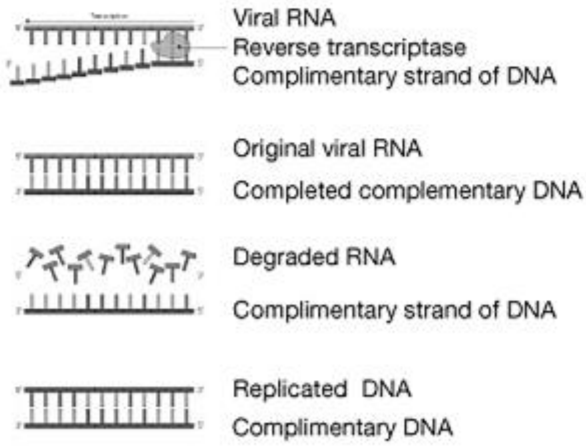


**Fig 1.6**

**(h)** Explain the role of the enzyme reverse transcriptase.

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For Examiner's use

Fig 1.7 shows how IgG antibodies against the SARS-CoV-2 virus work.

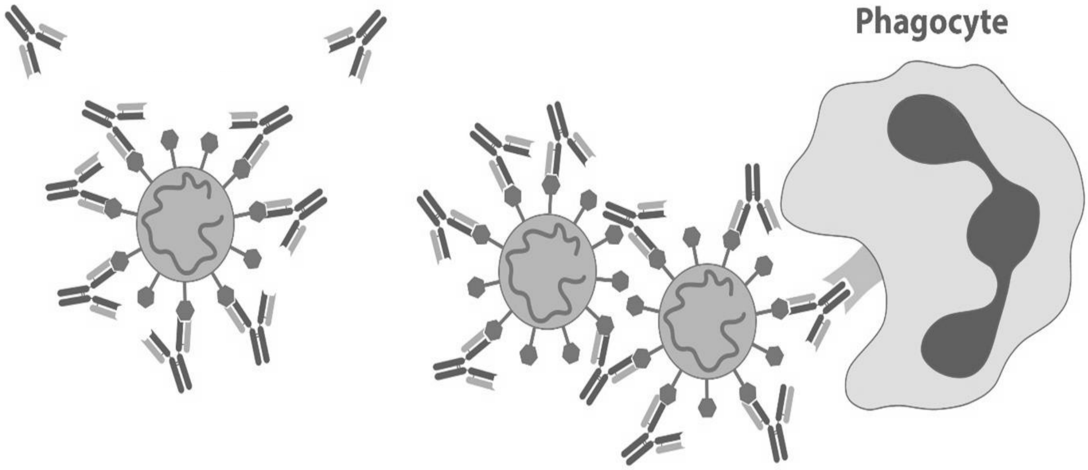


**Fig 1.7**

**(i)** With reference to Fig 1.7, explain how the structure of IgG facilitates its role in fighting off viral infections such as Covid-19.

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For Examiner's use

Many Covid-19 vaccines have been developed with differing efficacy. Fig 1.8 shows the various efficacy results obtained in phase-three trials of several Covid-19 vaccines.

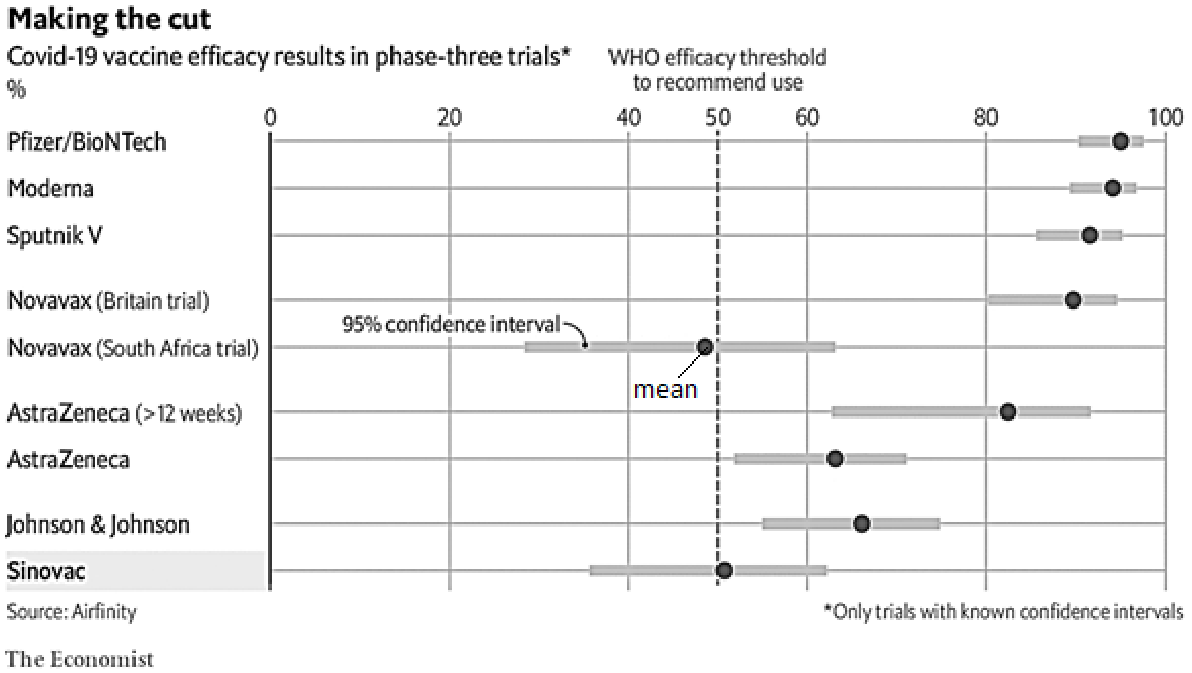


**Fig 1.8**

**(j)** With reference to Fig 1.8, describe how the data of the efficacy of the Pfizer vaccine and the Sinovac vaccine differ.

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[Total: 26]

For Examiner's use

## Question 2

Fig 2.1 depicts a model of cancer development from a single abnormal cell.

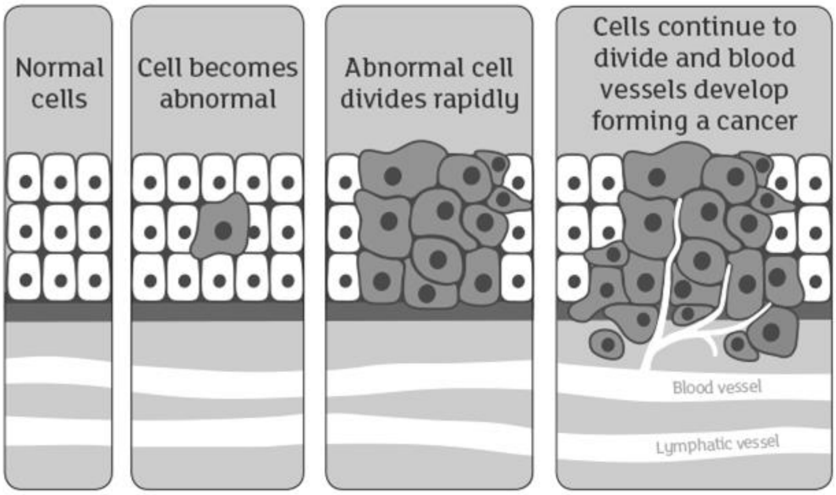


**Fig 2.1**

**(a)** Explain what may have led to the development of this abnormal cell from a normal cell before it divides rapidly.

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For Examiner's use

Retinoblastoma is a type of cancer that develops from a tumor in the retina, the tissue lining at the back of the eye. It is a result of mutations in the *Rb* gene. Fig 2.2 shows the different types Retinoblastoma and how they arise.

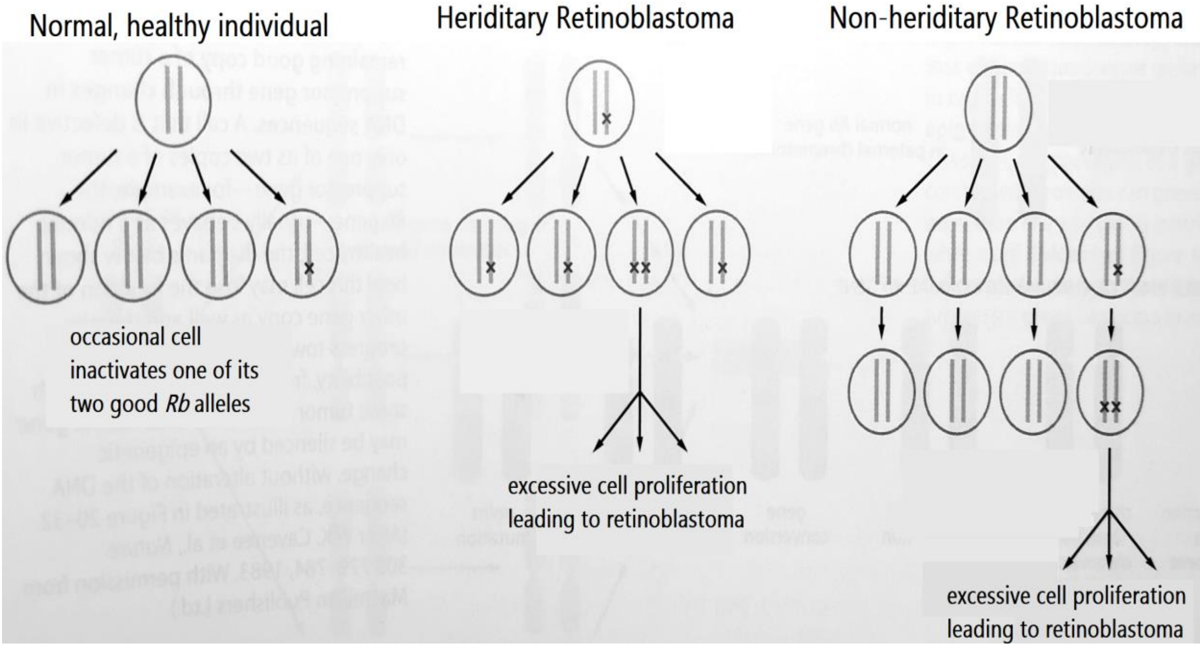


**Fig 2.2**

**(b)** Those with Hereditary Retinoblastoma usually suffer from more serious symptoms such as having multiple tumors in both eyes, while those who suffer from Non-hereditary Retinoblastoma usually develop only one tumor in one eye.

With reference to Fig 2.2, explain the observations.

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For Examiner's use

**(c)** One form of treatment for Retinoblastoma is chemotherapy, which involves the use of drugs to kill rapidly dividing cells. The drugs usually stimulate cell-signaling pathways, which lead to programmed cell death.

Retinoblastoma can come back a few years after chemotherapy, and usually, the new tumor formed is no longer sensitive to the same chemotherapeutic drugs.

Suggest how the drug resistant tumor has developed.

[3]

Recently, scientists discovered that some cells within tumors replace glucose with lipids as an energy source in order to multiply.

**(d)** **(i)** Lipids require more oxygen than glucose to be completely oxidized. Explain why.

[1]

**(ii)** Explain why it is better for tumor cells to oxidise lipids than glucose.

[2]

For Examiner's use

In another study investigating the differences between benign and malignant tumor cells, it was discovered that malignant tumor cells have cell surface membrane that is more fluid than benign cells.

**(e)** **(i)** State one possible structural difference between the cell surface membrane of malignant and benign tumor cells.

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**(ii)** Suggest an advantage of higher membrane fluidity in the transformation from benign to malignant tumor cells.

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[Total: 14]

For Examiner's use

## Question 3

Beta thalassemia is an inherited blood disorder caused by a 619bp deletion at the 3' end of the beta-globin gene. The beta-globin gene is 1.6kbp long and codes for beta subunits of haemoglobin. People suffering from beta thalassemia shows anemic symptoms.

Fig 3.1(a) shows the normal and mutated beta globin gene. To test for the presence of the mutation in individuals **W**, **X** and **Y**, the following steps were conducted.

1. DNA samples were extracted.
2. DNA samples were cut at the restriction sites indicated by (↑).
3. The fragments were separated by gel electrophoresis.

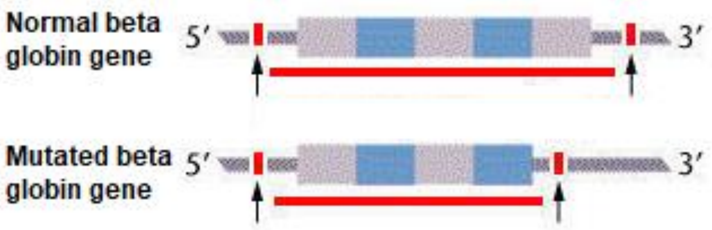


**Fig 3.1(a)**

Fig 3.1(b) shows the banding patterns on the gel obtained.

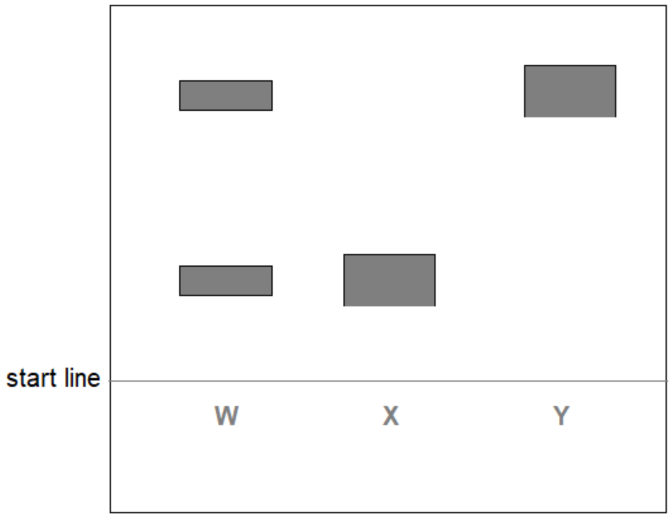


**Fig 3.1(b)**

*For Examiner's use*

**(a)** Explain how the 619bp deletion results in anemic symptoms.

[3]

**(b)** Identify which individual (**W**, **X** or **Y**)

**(i)** suffers from beta thalassemia: [1]

**(ii)** is a carrier of beta thalassemia: [1]

For Examiner's use

The treatment measures for beta thalassemia depend on the severity of the condition, and may include nutritional supplementation, blood transfusion and bone marrow transplant.

Another possible method is stem cell transplantation. Four genes, namely *Oct4, Kfl4, Sox2 and c-Myc*, encode transcription factors which could be used to reprogramme differentiated cells to induce pluripotency. Fig 3.2 shows the process of reprogramming somatic cells via (a) non-integrating plasmids and (b) retrovirus. The figure is not drawn to scale.

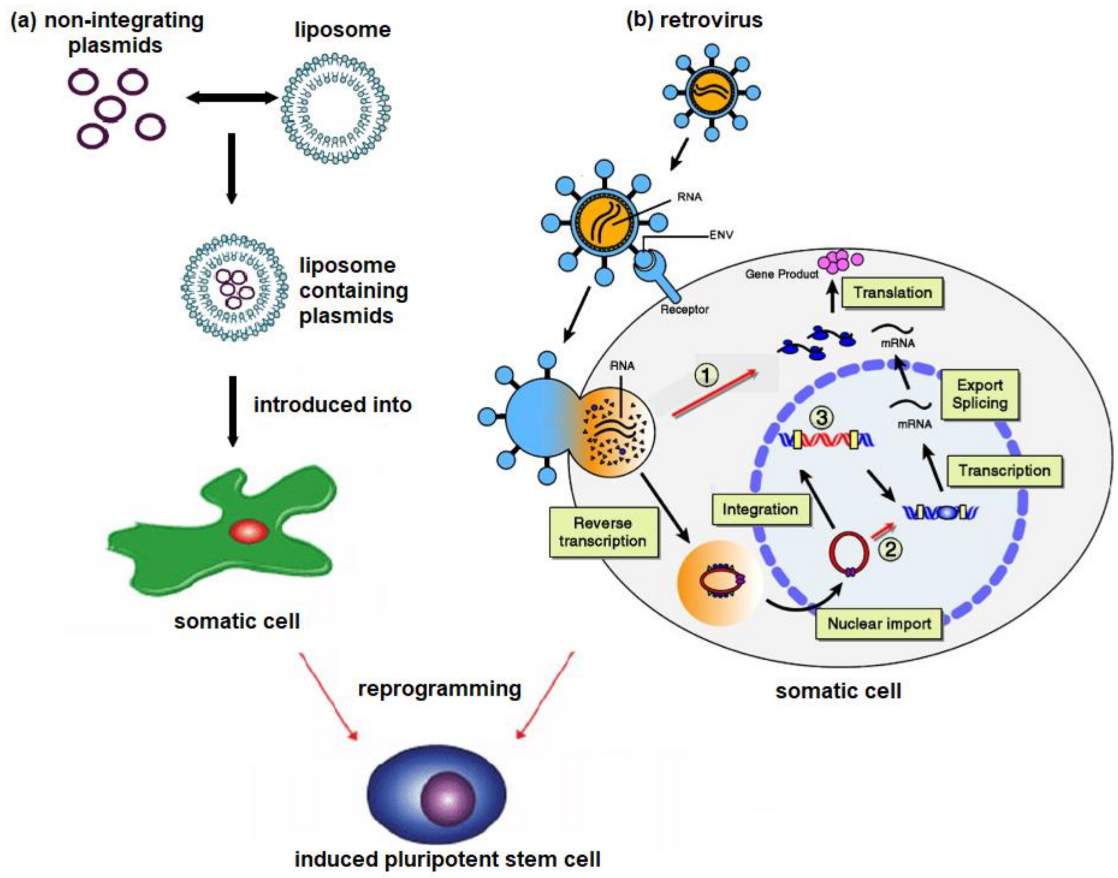


**Fig 3.2**

**(c)** Contrast the process of reprogramming somatic cells using non-integrating plasmids and retrovirus.

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*For Examiner's use*

**(d)** Another mutation to the beta globin gene gives proteins that are prone to polymerization under low oxygen condition. People who are homozygous with this mutated allele ($Hb^S$) suffer, and are likely to die, from sickle cell anaemia.

Malaria is a disease caused by the protozoan parasite, *Plasmodium*. It is found that in regions affected by malaria, the frequency of $Hb^S$ allele is higher than in regions not affected by malaria. This is because part of *Plasmodium*'s life cycle occurs within the red blood cells and the expression of $Hb^S$ results in sickle shaped red blood cells that are more easily hemolysed.

With climate change, the global temperature is expected to rise. Predict how the frequency of $Hb^S$ allele is likely to change in the temperate regions that are currently not affected by malaria.

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[Total: 10]

## Section B: Free-Response Questions (25 marks)

Answer only **ONE** question.
Write your answers on the answer booklet provided.
**Answer each part (a) and (b) on a fresh page.**

Your answers should be illustrated by large, clearly labelled diagrams, where appropriate.
Your answers must be in continuous prose, where appropriate.
Your answers must be set out in sections **(a)**, **(b)** etc., as indicated in the question.
A **NIL RETURN** is required.

### Question 4

**(a)** Outline the different levels of regulation of gene expression in eukaryotes and discuss their benefits. [15]

**(b)** Compare glucose synthesis and glucose oxidation. [10]

**Total: [25]**

**OR**

### Question 5

**(a)** Describe and explain the importance of membrane structure to its function. [15]

**(b)** Compare translation in eukaryotes and prokaryotes. [10]

**Total: [25]**

**DUNMAN HIGH SCHOOL**
**DHP YR 6**
**H2 BIOLOGY**

## Year 6 H2 Preliminary Exam 2021
## Suggested Answers

**<u>Paper 3</u>**
**Section A**

| **Question 1** | | |
|---|---|---|
| (a) | <u>Similarities</u><br>Both are enveloped viruses;<br>Both have RNA genome;<br>R! Both have glycoproteins / nucleocapsids<br><br><u>Differences</u><br>SARS-CoV-2 has positive sense RNA while influenza has negative sense RNA genome;<br>SARS-CoV-2 genome is a single piece of RNA while influenza has a segmented genome with 8 pieces of RNA;<br>SARS-CoV-2 has only 1 type of envelope glycoprotein (Spike) while influenza has 2 types of envelope glycoproteins (hemagglutinin and neuraminidase); | [4] |
| (b) | Obligate intracellular parasite – it requires a host cell in order to replicate;<br>Uses host cell's ribosomes to synthesize its enzymes (e.g. viral polymerase) and structural proteins e.g. N, S, M and E; (must name at least 1 viral protein)<br>Requires host Golgi to glycosylate its spike proteins;<br>Uses host Golgi membrane to form its envelope; (A: ER membrane)<br><br>Formed by assembly of pre-formed parts, does not grow and divide;<br>Assembly of nucleocapsid at Golgi;<br><br>4 max | [4] |
| (c) | It is a RNA virus, RNA polymerase / viral polymerase that is used to replicate its genome has no proof reading ability; | [1] |
| (d) | Mutations to the S1 gene  have resulted in changes amino acids at these positions (<u>T478K</u> and <u>L452R</u>);<br><br>These mutations could enhance the binding affinity of the spike protein to the ACE2 receptor;<br><br>thus enhancing viral adsorption / entry into host cells; | [3] |
| (e) | Comparing genome sequence between the different variants;<br>Larger the number of differences the more distantly related; | [2] |
| (f) | There is a lag time between infection and the production of antibodies; | [3] |

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|---|---|---|
| | so test could be negative when the person is having an active infection in this time period;<br><br>The test could be positive for antibodies after the person has already recovered from the infection as the antibodies can persist in the blood; | |
| (g) | Design of the sequence of the forward and reverse DNA primers;<br>complementary to sequences that are unique to the viral genome only; | [2] |
| (h) | Transcribe single stranded viral RNA genome into complementary double stranded DNA;<br><br>As PCR requires double stranded DNA as a template; | [2] |
| (i) | Neutralization – antigen binding sites of IgG complementary to viral spike proteins, binds to spike proteins such that the virus cannot attach to host receptors, thus preventing virus from entering cell;<br><br>Agglutination / aggregation. IgG has 2 antigen binding sites can bind to two viruses at the same time, thus viruses are held together / immobilized, and cannot move / disperse easily to infect cells;<br><br>Opsonisation: constant region provide recognition site for phagocyte leading to phagocytosis; | [3] |
| (j) | The Pfizer vaccine has a higher efficacy than the Sinovac vaccine (95%) vs (50%);<br><br>The Pfizer vaccine has a narrower 95% confidence interval than the Sinovac vaccine with range of less than 10% vs range of more than 25%; | [2] |

| Question 2 | | | |
|---|---|---|---|
| (a) | Loss of function mutation in tumour suppressor gene;<br><br>Gain of function mutation in proto-oncogene;<br><br>Accumulation of mutations in cell cycle checkpoint controls;<br><br>Any 2 | | [2] |
| (b) | *Rb* is a tumor suppressor gene;<br><br>In hereditary retinoblastoma, all cells started with one copy of mutant *rb* allele;<br><br>In the development of hereditary retinoblastoma, attaining one more mutation in the other *rb* allele is sufficient to turn normal cell into tumor cell;<br><br>higher chance of multiple cells attaining mutation; | | [4] |
| (c) | There exist variation in drug resistance within the population of tumor cells;<br><br>Due to high rate of cell division in tumor cells, mutation accumulates at higher rates;<br>OR<br>Ref to higher rate of errors in replication due to loss of DNA repair / proof-reading;<br><br>Drug selects for resistant cells, allowing them to continue dividing and passing on the allele to daughter cells; | | [3] |
| (d) | (i) | Lipids have more C-H bonds than glucose; | [1] |
| | (ii) | Oxidation of lipids produce two times more energy per gram;<br><br>To support the high rate of cell division which requires energy; | [2] |
| (e) | (i) | More cholesterol;<br>OR<br>More unsaturated fatty acids; | [1] |
| | (ii) | Enable the cell to squeeze endothelial layer into and out of blood vessel /AVP; | [1] |

| Question 3 | | |
|---|---|---|
| (a) | A large segment of beta-globin gene (39%) is deleted. This results in the synthesis of a non-functional beta subunit.;<br><br>Haemoglobin loses its specific quaternary structure;<br><br>hence it cannot transport oxygen efficiently.; | [3] |
| (b) | (i) Y;<br><br>(ii) W; | [1]<br><br>[1] |
| (c) | Retrovirus adsorbs to specific host receptors before fusing with cell membrane, while liposome fuses directly with cell membrane.;<br><br>Retrovirus carries RNA into the cell while liposome carries DNA;<br><br>Genes carried by retrovirus have to be integrated into the chromosome before it can be expressed while genes carried by the liposomes can be directly expressed;<br><br>Max 3 | [2] |
| (d) | Prediction: $Hb^S$ allele frequency increase;<br><br>Malaria transmitted by *Anopheles* mosquito vector + Migration of mosquitos to temperate regions where currently was too cold, spreading malaria;<br><br>Poses selection pressure to select for $Hb^S$ allele; | [3] |

**Section B**
**Question 4**
**(a)** Outline the different levels of regulation of gene expression in eukaryotes and discuss their benefits. [15]

**<u>Different levels of regulation of gene expression (max 12)</u>**
<u>Chromatin level</u>

1. Ref to DNA methylation and histone deacetylation increasing condensation of chromatin / ORA;
2. Ref to more positive charges on histone tails hence higher affinity for negatively charged DNA / ORA;
3. Densely packed heterochromatin prevents access of general transcription factors and RNA polymerase to the promoter, reducing the rate of transcription / ORA;

<u>Transcriptional level</u>

4. General <u>transcription factors</u> binding to the TATA box recruits RNA pol to the promoter and initiate transcription;
5. <u>Activator</u> proteins binding to <u>enhancer</u> sequence stabilises the transcription initiation complex on the promoter + hence increasing the rate of transcription;
   OR
   <u>Repressor</u> proteins binding to the <u>silencer</u> sequence destabilises the transcription initiation complex on the promoter + hence inhibiting transcription;

<u>Post-transcriptional level</u>

6. 5' capping of mRNA tags the mRNA for export to the cytoplasm for translation;
7. Excision of introns and alternative splicing of exons together enables different protein products to be coded for by one gene;

<u>Translational</u>

8. Length of polyA tail determines the half life of the mRNA / AW and hence the amount of product that can be translated from the mRNA;
9. Binding of translation initiation factors to the 5' end of the mRNA facilitates the recruitment of ribosomes;
   OR
   Binding of regulatory proteins can prevent the binding of ribosomes;

<u>Post-translational</u>

10. Chemical modification of proteins (eg. [any 1] glycosylation/phosphorylation) activates / deactivates the protein;
11. Structural modification (eg [any 1] cleaving / removal of specific number of amino acids / addition of disulphide linkages) to give functional protein;
12. Tagging proteins with ubiquitin / polyubiquitination for proteasome degradation;

**<u>Benefits of different levels of regulation (max 5)</u>**

13. – 14. Ref. to different functions of regulation;;
Eg. chromatin level regulation occurs during differentiation of cell **+ compared to** transcriptional regulation occurs during cell response to environmental changes.

    **note**: marks awarded for well elaborated benefit of each level

15. – 16. Ref. to different **scale of regulation**;;
Eg. chromatin level of regulation results in large number of genes silenced / expressed **+ compared to** transcriptional that only upregulate / silence one or few genes.

17. Enable **coordinated control** of gene expression OR enable genes of related functions to be turned on or off at the same time;
18. Ref to these genes having the **same distal control element**;

19. Translational level of regulation allows amount of proteins to be regulated to synthesize only what is needed and minimise wastage of resources;

20. Post-translational level of protein modification enable **quick regulation** of protein activity;

**Total max 14**

**4(b)** Compare glucose synthesis and glucose oxidation. [10]

**Differences**

1. Glucose is synthesized after **photosynthesis** while it is oxidized during **respiration**;

2. Glucose synthesis **requires $CO_2$** while complete glucose oxidation **releases $CO_2$**;

3. Glucose synthesis **releases $O_2$** while complete glucose oxidation **requires $O_2$**;

4. **Water is photolysed** during the process of glucose synthesis while water is synthesized from the complete **oxidation of glucose**;

5. Glucose synthesis is driven by **light energy** while glucose oxidation is driven by **chemical energy**;

6. Glucose synthesis takes place in **photosynthetic organisms (eg plants) / within chloroplast** while glucose oxidation takes place in **most organisms / requires mitochondria** ;

**Similarities**

7. Both process requires a **series enzymatic reactions** / both requires enzymes ;

8. Both processes involves the **utilization of ATP**;

9. Both processes involves the **reduction and oxidation of coenzymes**;

10. In both processes, **ATP is synthesized via chemiosmosis**;

**Question 5**
(a) Describe and explain the importance of membrane structure to its function. [15]

| **Structure of membrane** | **Importance in membrane function** |
|---|---|
| 1. Comprise of **phospholipids**;<br><br>2. Ref to **amphipathic nature of phospholipids** + **bilayer** ; | 3. Enables only **small and non-polar** substance to move across<br>OR<br>prevents large and/or polar substance from moving across the membrane;<br><br>1. Enables **compartmentalisation** within the cell;<br>2. Any relevant example;<br>Eg prevents hydrolytic enzymes in lysosomes to digest the cell contents |
| 3. Ref to **C=C double bonds of fatty acid chains** that results in **kinks**;<br><br>4. Contains **cholesterol**; | 5. Ref to **fluidity** of the membrane;<br>6. Relevant example on how fluidity enables a membrane function;<br>Eg. enables the formation of vesicles during endocytosis |
| 7. Contains **transmembrane / integral proteins** ; | 8. May function as **transport proteins** that enables **facilitated diffusion and active transport**;<br><br>9. Function as **enzymes** required at the membrane;<br>10. Relevant example;<br>Eg. $NADP^+$ reductase in photosynthesis<br><br>11. Function as **receptors** that **receives extracellular signals** for cell-cell signalling (important since membrane poses as a barrier for direct communication);<br>12. Relevant example;<br>Eg. Receptor tyrosine kinase that receives insulin / G-protein coupled receptor that receive glucagon; |

| | |
|---|---|
| 13. Contains **peripheral proteins** ; | 14. **Transduce / relay extracellular signal** for cell-cell signalling;<br>15. Relevant example;<br>Eg. G protein activating adenylyl cyclase |
| 16. Contains **glycoproteins and glycolipids** + mention that this is only present in **cell surface membrane** and **only on the extracellular face**; | 17. **Receptors** in cell-cell communication;<br>**Note: point 11 and 17 award once**<br>18. Relevant example;<br>Eg. MHC class II on professional APC are glycoproteins that present antigens to $CD4^+$ T cells, activating the adaptive immune response;<br><br>19. Required in **cell-cell adhesion** in the **formation of tissues**; |

**(b)** Compare translation in eukaryotes and prokaryotes. [10]

**Differences:**

| No | Point of Comparison | Eukaryotes | Prokaryotes |
|---|---|---|---|
| 1 | mRNA | **Monocistronic mRNA** which codes for 1 polypeptide | **Polycistronic mRNA** which codes for 1 polypeptide which will be cleaved into several functional polypeptides of a particular metabolic pathway |
| 2 | Location of translation | **Free ribosomes in cytoplasm and ribosomes bound to rough endoplasmic reticulum** | **Free ribosomes in cytoplasm only** |
| 3 | Ribosomes | Uses **80S ribosomes** (40S small subunit + 50S large subunit) for translation | Uses **70S ribosomes** (30S small subunit + 50S large subunit) for translation |
| 4 | Amino acid carried by initiator tRNA | Initiator tRNA carries **methionine** | Initiator tRNA carries **formyl-methionine** |

**Similarities:**

5. In both prokaryotes and eukaryotes, mRNA is **synthesized from 5' → 3'** and polypeptides are synthesized from N-terminal to C-terminal / DNA is **read from 3' → 5'** during transcription and mRNA is **read from 5' → 3'** during translation.;

6. Translation in both prokaryotes and eukaryotes involves activation of amino acids by aminoacyl-tRNA synthetases, which forms a covalent bond between amino acid and its corresponding tRNA.;

7. Codon **AUG** acts as signals to start translation and codons **UAG, UAA and UGA** act as signals to stop translation in both prokaryotes and eukaryotes.;

8. Translation termination in both prokaryotes and eukaryotes involves additional of **release factor** when stop codon on mRNA reaches the 'A' site.

9. Multiple ribosomes can **translate a single mRNA simultaneously** (polyribosomes) in both prokaryotes and eukaryotes.;

10. AVP